

Interpretable Latent Representations of Neural Activity via Sparse Autoencoders

GitHub:



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ABSTRACT

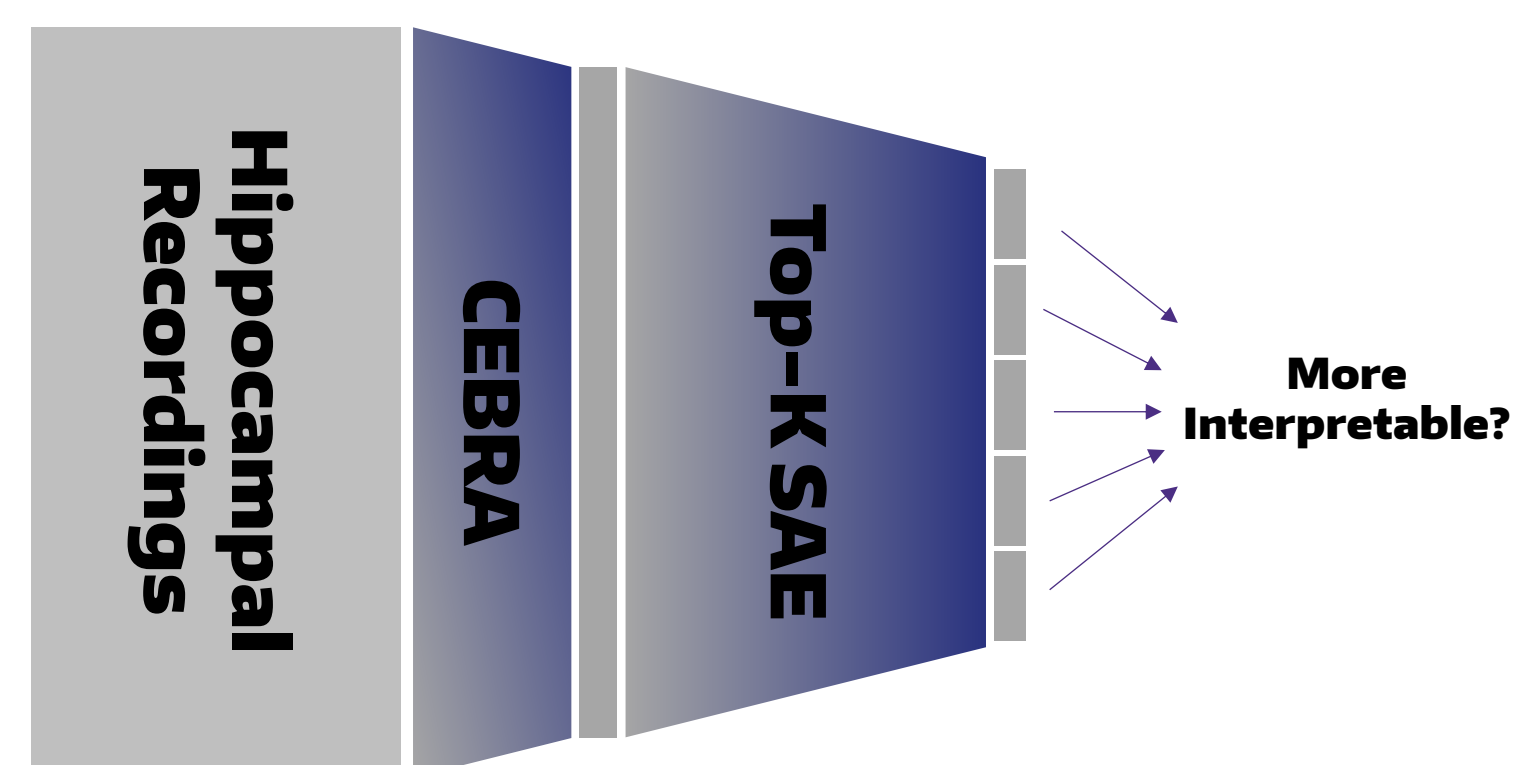
We post-hoc sparsify CEBRA neural embeddings with Top-k sparse autoencoders (SAEs) to make latent dimensions attributable to specific neurons and behavioral variables. Sparsification activates fewer dimensions per sample, concentrates attribution, and preserves (often slightly improves) position and direction decoding. Latent knockout tests show a small subset of SAE dimensions accounts for most decoding loss, localizing responsibility within the embedding.

BACKGROUND

- Modern latent models like CEBRA decode behavior well but pack information into many entangled dimensions, making it unclear which neurons or variables drive each axis.
- To test hypotheses or design perturbations, we want latents that map cleanly to small neuron sets or specific behavioral factors, and whose removal has predictable effects.
- Sparse autoencoders have been used to make vision, language, and gene embeddings more interpretable by forcing localized, sparse features. We adapt that post-hoc trick to neural population embeddings.
- Goal:** Use SAEs to turn dense embeddings into sparse, attributable ones

Definitions

- Working definition (for this poster):** A latent dimension is **interpretable** if we can **attribute** its activity to a compact set of neurons and/or a specific behavioral variable, and if perturbing that dimension (or its key neurons) produces a predictable, specific change in decoding.
- Latent dimension:** one axis of the embedding space.
- Attribution:** strength of neuron/behavior $\rightarrow$ latent relationship
- Sparsity:** % of latent units active per sample.
- CEBRA:** CEBRA is a contrastive learning method that maps neural population activity into low-dimensional embeddings aligned with behavior.



Workflow

Methods

- Data.** Four single-session hippocampal tetrode recordings (Achilles, Buddy, Cicero, Gatsby), 25 ms bins. Behaviors: 1D track position and running direction (L/R). Spikes were z-scored; behavior aligned to bins.
- CEBRA.** Supervised, causal temporal convs; latent dims = 8, 16, 32, 64. We selected the best decoding config as the dense baseline.
- SAE (post-hoc).** Overcomplete ($\sim 8\times$) encoder/decoder with **Top-k** nonlinearity ($k \in \{8, 16, 32\}$); 50 epochs, LR=1e-3, batch=256; loss = MSE(Z, Z).
- Metrics.** Sparsity-explained variance curve: non-zero ratio per sample vs. reconstruction R2

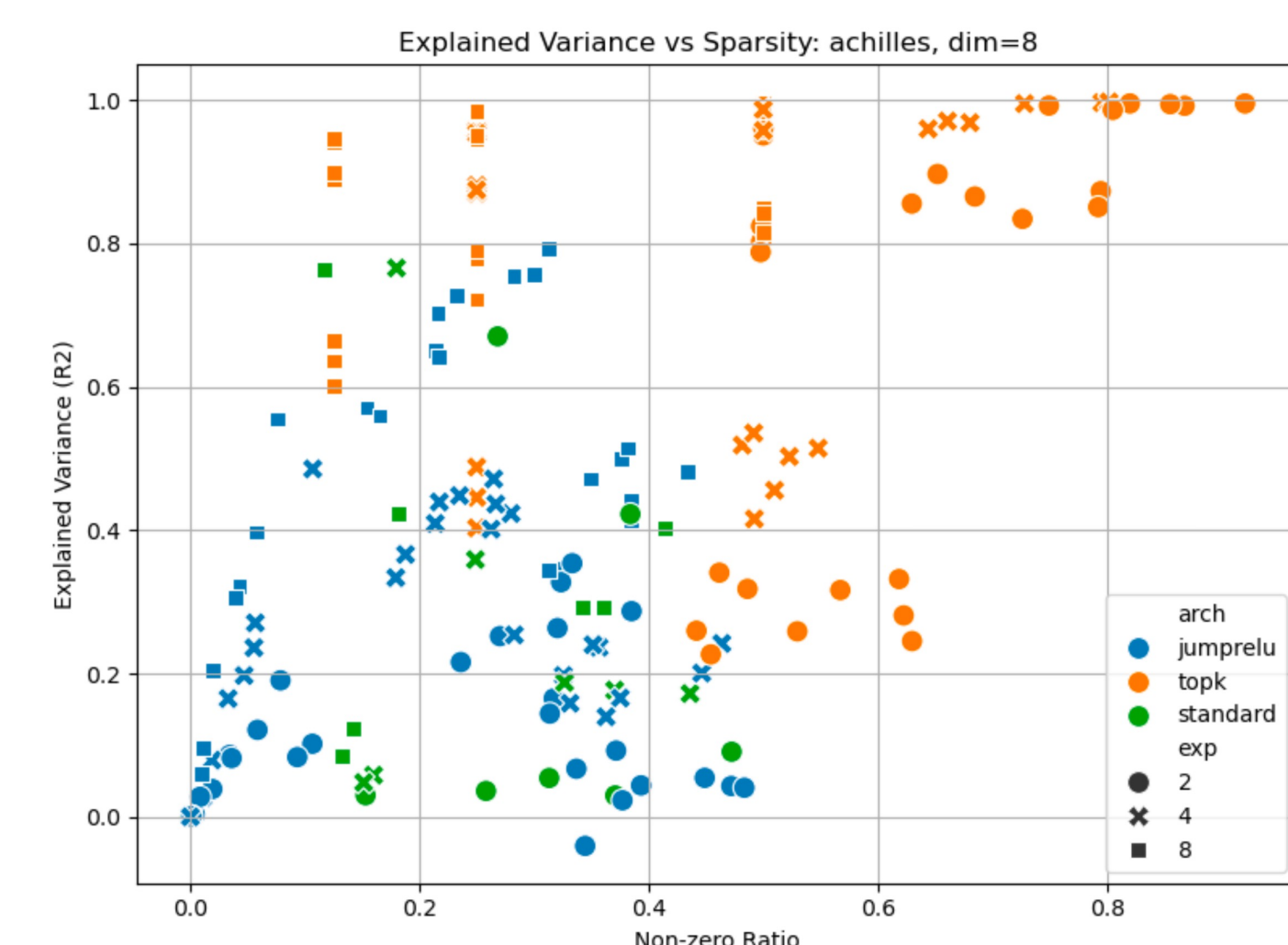


Fig 2. TopK SAE in general had the highest reconstruction fidelity at a high sparsity. One CEBRA model example (Achilles, dim = 8) shown.

Attribution Pre vs Post SAE

Decoding (utility control):

- Position:** Fit a ridge regressor on the full latent matrix (CEBRA or SAE) to predict 1D position; report validation R^2 .
- Direction:** Fit a logistic classifier to predict L/R; report accuracy (and AUROC in appendix).
- Same train/val split and regularization grid for all models to ensure fair comparison.

Neuron-latent attribution score:

- Compute absolute Pearson correlation between each neuron's activity and each latent over time.
- For each latent z_j , take the top-10 absolute correlations and average them.
- Report the mean of that value across latents = "Neuron Score".

Behavior-latent attribution score:

- For each latent z_j , fit a ridge regressor to each behavioral component (position, left, right) and take the best R^2 .
- Average over latents = "Behavior Score".

Evidence of no loss in utility:

- Compare decoding metrics (R^2 , direction accuracy) pre vs post SAE.
- Plot bars with CEBRA vs SAE attribution scores rise, decoding stays flat/slightly improves.

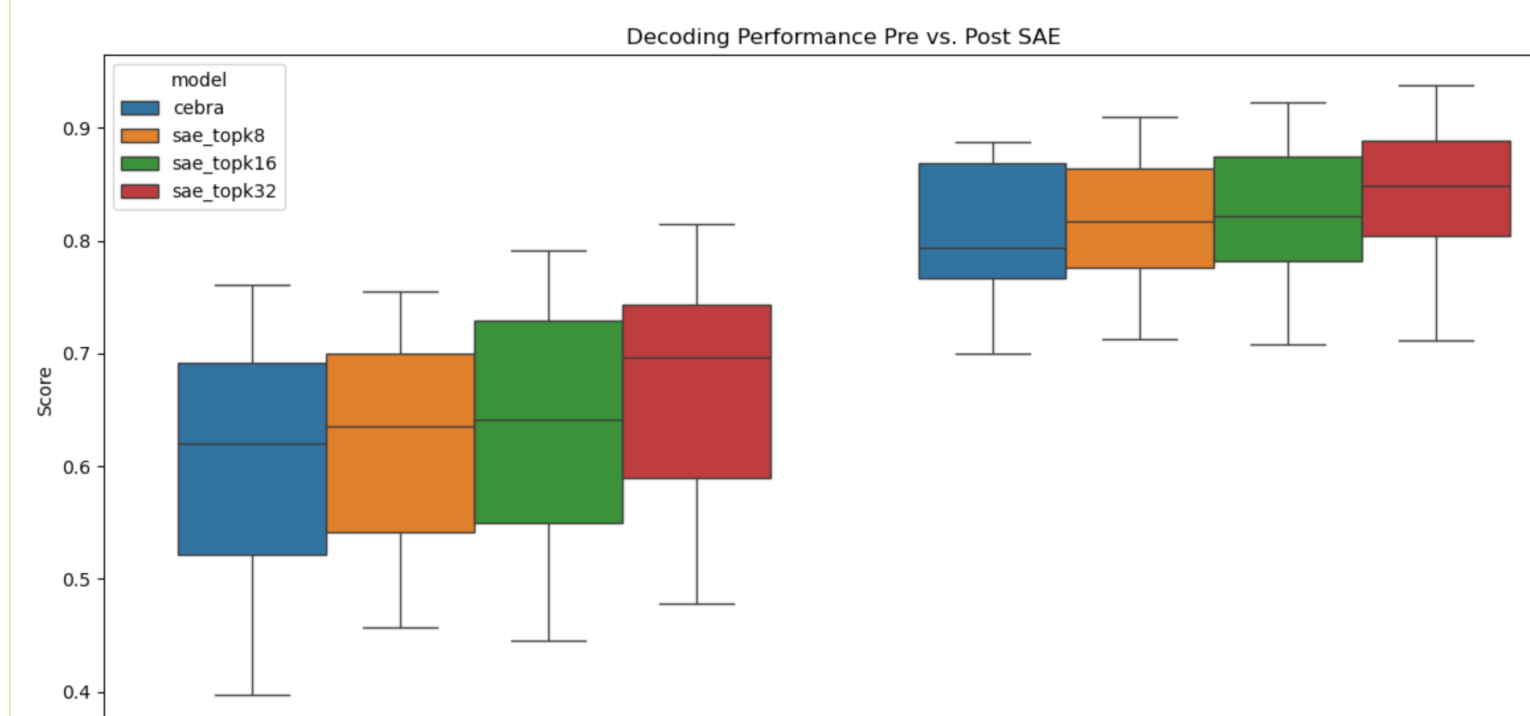


Fig 3. Top-k SAE preserves accuracy: no significant drop in either behavioral decoding (position, direction) or neuron-level reconstruction.

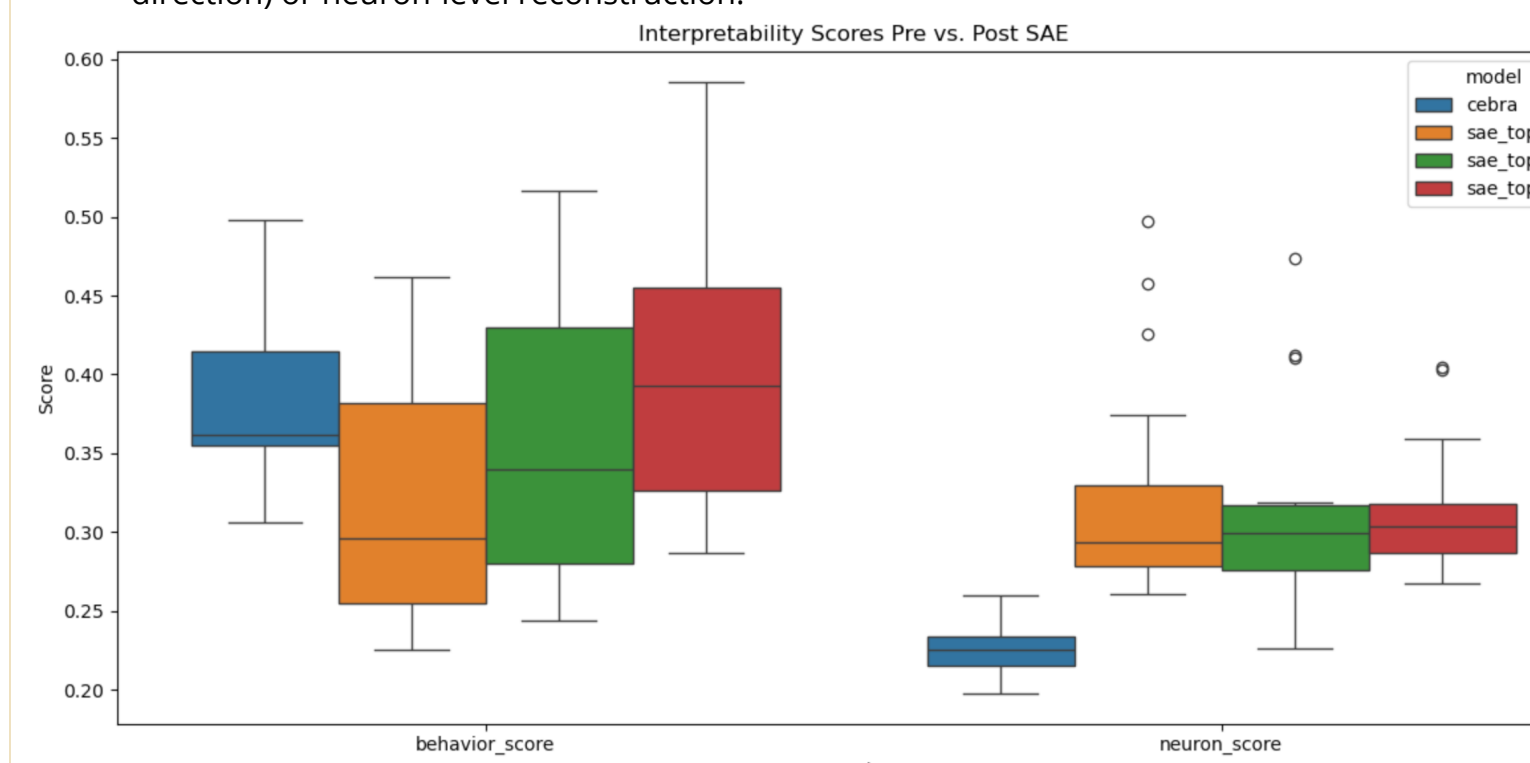


Fig 4. Sparse SAE markedly raises neuron-latent attribution, while behavior-latent attribution remains comparable to the dense CEBRA baseline.

Latent-Axis Knockout (Responsibility Test)

- What we do:** We first train a Ridge regression model on the full set of embeddings to predict the animal's 1D position. Then, for each latent dimension, we zero out that column and re-evaluate the decoder.
- How we measure:** The drop in ΔR^2 between the baseline and "knockout" run quantifies each axis's unique contribution to decoding accuracy.
- Why it matters:** A few axes with large ΔR^2 indicate that the model's representational power is concentrated. SAEs, especially TopKSAE, channel positional information into fewer, high-responsibility dimensions, making them more interpretable.

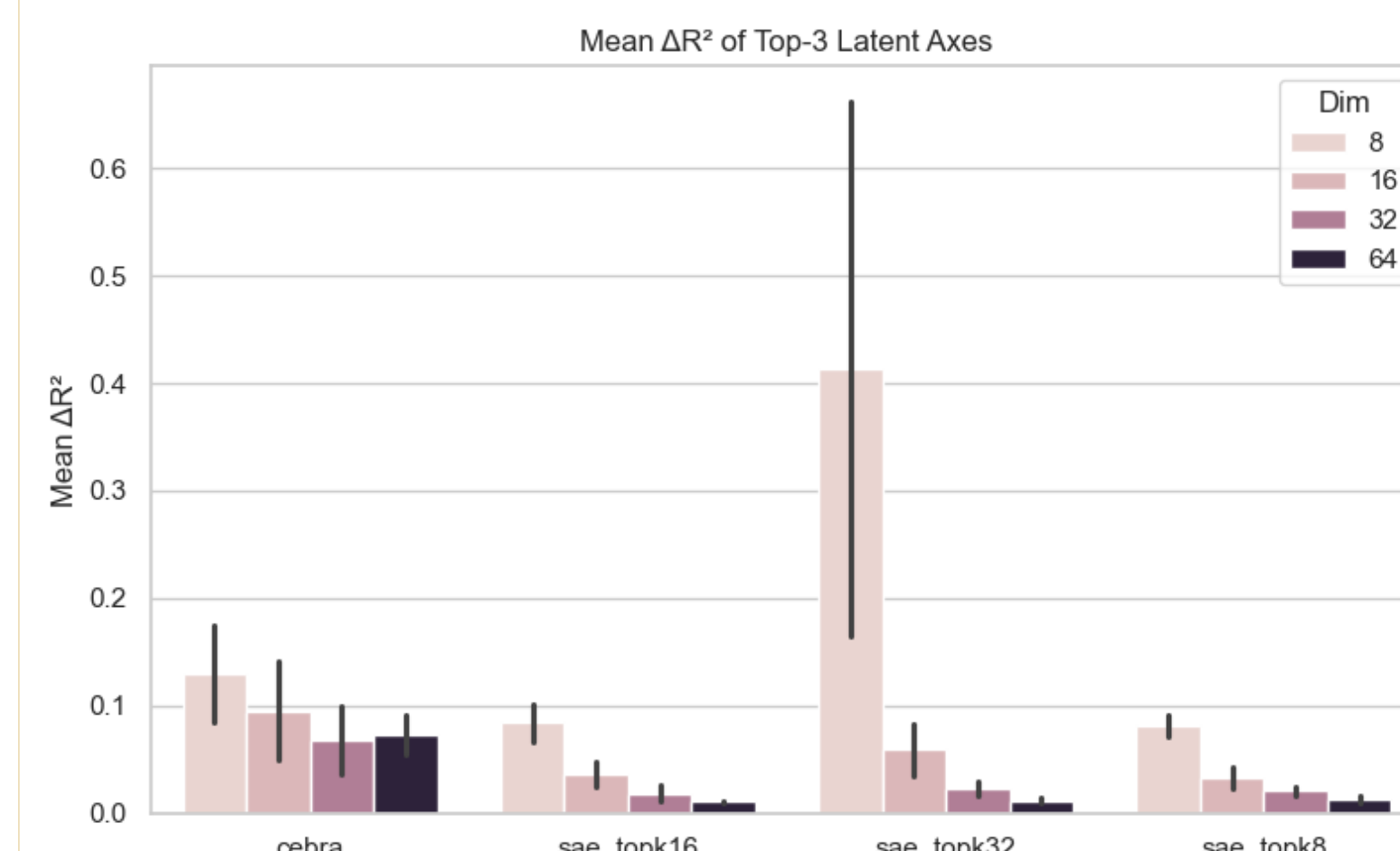


Fig 5. SAE_topk32 drives the largest mean drop in R^2 among its top three axes, indicating it focuses positional information into far fewer, more critical dimensions than CEBRA.

Discussion

What we showed

- Top-k sparse autoencoders make CEBRA's dense embeddings much sparser **without hurting decoding**.
- Sparse latents line up more cleanly with individual neurons and behaviors.
- Knocking out one latent at a time confirms that only a few dimensions drive most of the decoding power.

Main caveats

- Correlations and knockouts don't prove brain mechanisms.
- We haven't linked sparse latents to place cells or anatomy yet.
- Results come from one dataset and one sparsifier; broader tests are still needed.

Next steps:

- Compare to other sparsity methods (e.g., NMF, varimax).
- Check stability across random seeds and recording sessions.
- Map high-loading neurons to known tuning (place fields) and brain layers.